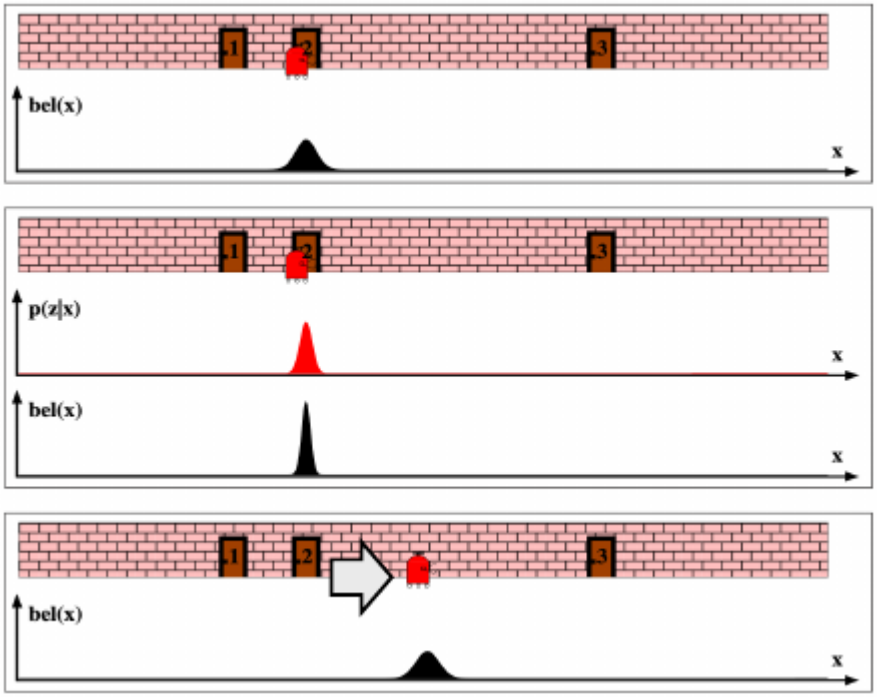
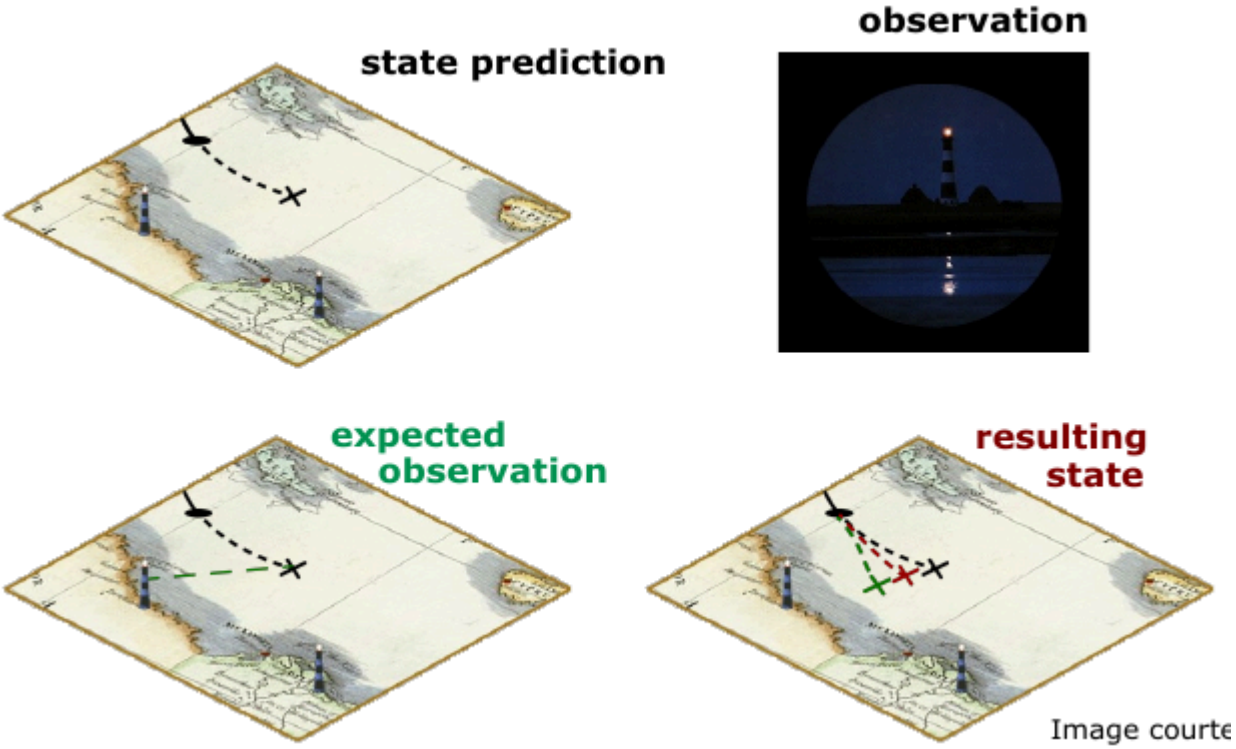
Ro10: EKF Localisation



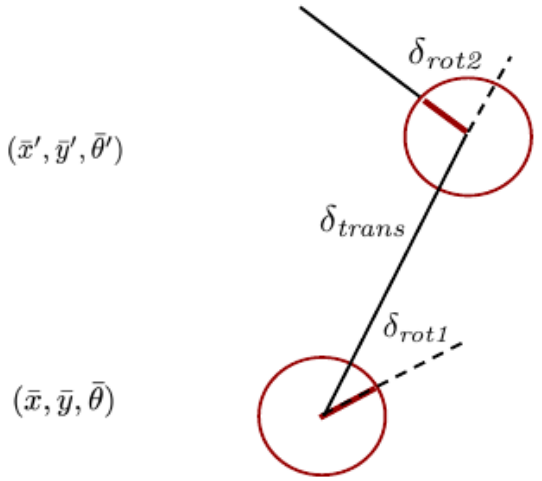


EKF Localisation for Landmark Maps

$$
\begin{aligned}
&\textbf{Extended_Kalman_filter}(\mu_{t-1}, \Sigma_{t-1}, u_t, z_t)\textbf{:} \\
&\quad \bar{\mu}_t = \underline{g(u_t, \mu_{t-1})} \\
&\quad \bar{\Sigma}_t = \underline{G_t}\, \Sigma_{t-1}\, G_t^T + \underline{R_t} \\
&\quad K_t = \bar{\Sigma}_t\, H_t^T (H_t\, \bar{\Sigma}_t\, \underline{H_t^T} + \underline{Q_t})^{-1} \\
&\quad \mu_t = \bar{\mu}_t + K_t(z_t - \underline{h}(\bar{\mu}_t)) \\
&\quad \Sigma_t = (I - K_t\, H_t)\, \bar{\Sigma}_t \\
&\quad \text{return } \mu_t, \Sigma_t
\end{aligned}
$$

Odometry Model

- Motion from $(\overline{x}, \overline{x}, \overline{\theta})$ to $(\overline{x}', \overline{y}', \overline{\theta}')$ with $u = (\delta rot1, \delta trans, \delta rot2)$



$$
\begin{aligned}
\delta_{trans} &= \sqrt{(\bar{x}' - \bar{x})^2 + (\bar{y}' - \bar{y})^2} \\
\delta_{rot1} &= \operatorname{atan2}(\bar{y}' - \bar{y}, \bar{x}' - \bar{x}) - \bar{\theta} \\
\delta_{rot2} &= \bar{\theta}' - \bar{\theta} - \delta_{rot1}
\end{aligned}
$$

Gaussian noise in control space

$$u \sim \mathcal{N}(0, \Sigma)$$

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Motion Model

- Odometry motion update

$$\begin{pmatrix} x' \\ y' \\ \theta' \end{pmatrix} = \underbrace{\begin{pmatrix} x \\ y \\ \theta \end{pmatrix} + \begin{pmatrix} \delta_{\text{trans}} \cos(\theta + \delta_{rot_1}) \\ \delta_{\text{trans}} \sin(\theta + \delta_{rot_1}) \\ \delta_{rot_1} + \delta_{rot_2} \end{pmatrix}}_{g(u_t, x_{t-1})} + \mathcal{N}(0, R_t)$$

Non-linear model

Noise in state space

- Linearization

$$g(u_t, x_{t-1}) \approx g(u_t, \mu_{t-1}) + G_t(x_{t-1} - \mu_{t-1})$$

Jacobian at μ_{t-1}

-
- **Jacobian**
  - Compute Jacobian for $\begin{pmatrix} x' \\ y' \\ \theta' \end{pmatrix} = \begin{pmatrix} x + \delta_{\text{trans}} \cos(\theta + \delta_{rot_1}) \\ y + \delta_{\text{trans}} \sin(\theta + \delta_{rot_1}) \\ \theta + \delta_{rot_1} + \delta_{rot_2} \end{pmatrix}$
  - which is

$$G_t = \frac{\partial g(u, \mu_{t-1})}{\partial x_{t-1}} = \frac{\partial g(u, x)}{\partial x} = \begin{pmatrix} \frac{\partial x'}{\partial x} & \frac{\partial x'}{\partial y} & \frac{\partial x'}{\partial \theta} \\ \frac{\partial y'}{\partial x} & \frac{\partial y'}{\partial y} & \frac{\partial y'}{\partial \theta} \\ \frac{\partial \theta'}{\partial x} & \frac{\partial \theta'}{\partial y} & \frac{\partial \theta'}{\partial \theta} \end{pmatrix}$$

$$G_t = \begin{pmatrix} 1 & 0 & -\delta_{\text{trans}} \sin(\theta + \delta_{rot_1}) \\ 0 & 1 & \delta_{\text{trans}} \cos(\theta + \delta_{rot_1}) \\ 0 & 0 & 1 \end{pmatrix}$$

  -
- **Noise**
  - EKF requires noise in state space
  - $x_t = b(u_t, x_{t-1}) + \mathcal{N}(0, R_t)$ Noise in state space
  - Odometry noise $\mathcal{N}(0, M_t)$ specified in control space $u = (\delta rot1, \delta trans, \delta rot2)$
  - Mapping based on 1st order error propagation
    - $R_t = V_t M_t V_t^T$ with $V_t = \frac{\partial g(u,x)}{\partial u}$

Observation Model

- Range-Bearing Model

$$z_t^i = \begin{bmatrix} r_t^i \\ \phi_t^i \end{bmatrix} = \begin{bmatrix} \sqrt{(m_{j,x}-x)^2 + (m_{j,y}-y)^2} \\ atan2(m_{j,y}-y, m_{j,x}-x) - \theta) \end{bmatrix} + (0, Q_t)$$

- Linearisation : $h(x_t, m) \approx h(\overline{\mu}_t) + H_t^i(x_t - \overline{\mu}_t)$
- **Jacobian**
  - Compute Jacobian of $\begin{pmatrix} r_t^i \\ \phi_t^i \end{pmatrix} = \begin{pmatrix} \sqrt{(m_{j,x}-x)^2 + (m_{j,y}-y)^2} \\ \text{atan2}(m_{j,y}-y, m_{j,x}-x) - \theta \end{pmatrix}$
  - Leads to

$$H_t^i = \frac{\partial h(x,m)}{\partial x} = \begin{pmatrix} \frac{\partial r_t^i}{\partial x} & \frac{\partial r_t^i}{\partial y} & \frac{\partial r_t^i}{\partial \theta} \\ \frac{\partial \phi_t^i}{\partial x} & \frac{\partial \phi_t^i}{\partial y} & \frac{\partial \phi_t^i}{\partial \theta} \end{pmatrix}$$

$$H_t^i = \begin{pmatrix} -\frac{m_{j,x}-x}{\sqrt{q}} & -\frac{m_{j,y}-y}{\sqrt{q}} & 0 \\ \frac{m_{j,y}-y}{q} & -\frac{m_{j,x}-x}{q} & -1 \end{pmatrix}$$

$$\text{with} \quad q = (m_{j,x}-x)^2 + (m_{j,y}-y)^2$$

EKF

Extended_Kalman_filter$(\mu_{t-1}, \Sigma_{t-1}, u_t, z_t)$**:**

$\bar{\mu}_t = g(u_t, \mu_{t-1})$
$\bar{\Sigma}_t = G_t \, \Sigma_{t-1} \, G_t^T + R_t$

Prediction

$K_t = \bar{\Sigma}_t \, H_t^T (H_t \, \bar{\Sigma}_t \, H_t^T + Q_t)^{-1}$
$\mu_t = \bar{\mu}_t + K_t(z_t - h(\bar{\mu}_t))$
$\Sigma_t = (I - K_t \, H_t) \, \bar{\Sigma}_t$
$return \; \mu_t, \Sigma_t$

Correction

1. **Algorithm EKF localization** $(\mu_{t-1}, \Sigma_{t-1}, u_t, z_t, c_t, m)$:

2. $\theta = \mu_{t-1,\theta}$

3. $G_t = \begin{pmatrix} 1 & 0 & -\delta_{\text{trans}} \sin(\theta + \delta_{\text{rot}_1}) \\ 0 & 1 & \delta_{\text{trans}} \cos(\theta + \delta_{\text{rot}_1}) \\ 0 & 0 & 1 \end{pmatrix}$ ← **Jacobian of g** $G = \frac{\partial g}{\partial x}$

4. $V_t = \begin{pmatrix} -\delta_{\text{trans}} \sin(\theta + \delta_{\text{rot}_1}) & \cos(\theta + \delta_{\text{rot}_1}) & 0 \\ \delta_{\text{trans}} \cos(\theta + \delta_{\text{rot}_1}) & \sin(\theta + \delta_{\text{rot}_1}) & 0 \\ 1 & 0 & 1 \end{pmatrix}$ ← **Control space to state space** $V = \frac{\partial g}{\partial u}$

5. $M_t = \begin{pmatrix} \alpha_1 \delta_{\text{rot}_1}^2 + \alpha_2 \delta_{\text{trans}}^2 & 0 & 0 \\ 0 & \alpha_3 \delta_{\text{trans}}^2 + \alpha_4 (\delta_{\text{rot}_1}^2 + \delta_{\text{rot}_2}^2) & 0 \\ 0 & 0 & \alpha_1 \delta_{\text{rot}_2}^2 + \alpha_2 \delta_{\text{trans}}^2 \end{pmatrix}$

6. $\bar{\mu}_t = \mu_{t-1} + \begin{pmatrix} \delta_{\text{trans}} \cos(\theta + \delta_{\text{rot}_1}) \\ \delta_{\text{trans}} \sin(\theta + \delta_{\text{rot}_1}) \\ \delta_{\text{rot}_1} + \delta_{\text{rot}_2} \end{pmatrix}$ ← **Prediction using g**

7. $\bar{\Sigma}_t = G_t \Sigma_{t-1} G_t^T + V_t M_t V_t^T$ ← **Uncertainty of the prediction**

8: $Q_t = \begin{pmatrix} {\sigma_r}^2 & 0 \\ 0 & {\sigma_\phi}^2 \end{pmatrix}$

9: **for** all observed features $z_t^i = (r_t^i, \phi_t^i)^T$ **do**

10: $\quad j = c_t^i$

11: $\quad q = (m_{j,x} - \bar{\mu}_{t,x})^2 + (m_{j,y} - \bar{\mu}_{t,y})^2$

12: $\quad \hat{z}_t^i = \begin{pmatrix} \sqrt{q} \\ \text{atan2}(m_{j,y} - \bar{\mu}_{t,y}, m_{j,x} - \bar{\mu}_{t,x}) - \bar{\mu}_{t,\theta} \end{pmatrix}$ ← **Expected measurement**

13: $\quad H_t^i = \begin{pmatrix} -\frac{m_{j,x} - \bar{\mu}_{t,x}}{\sqrt{q}} & -\frac{m_{j,y} - \bar{\mu}_{t,y}}{\sqrt{q}} & 0 \\ \frac{m_{j,y} - \bar{\mu}_{t,y}}{q} & -\frac{m_{j,x} - \bar{\mu}_{t,x}}{q} & -1 \end{pmatrix}$ ← **Jacobian of h**

14: $\quad S_t^i = H_t^i \bar{\Sigma}_t {H_t^i}^T + Q_t$

15: $\quad K_t^i = \bar{\Sigma}_t [H_t^i]^T [S_t^i]^{-1}$

16: $\quad \bar{\mu}_t = \bar{\mu}_t + K_t^i (z_t - \hat{z}_t^i)$

17: $\quad \bar{\Sigma}_t = (I - K_t^i H_t^i) \bar{\Sigma}_t$

18: **end for**

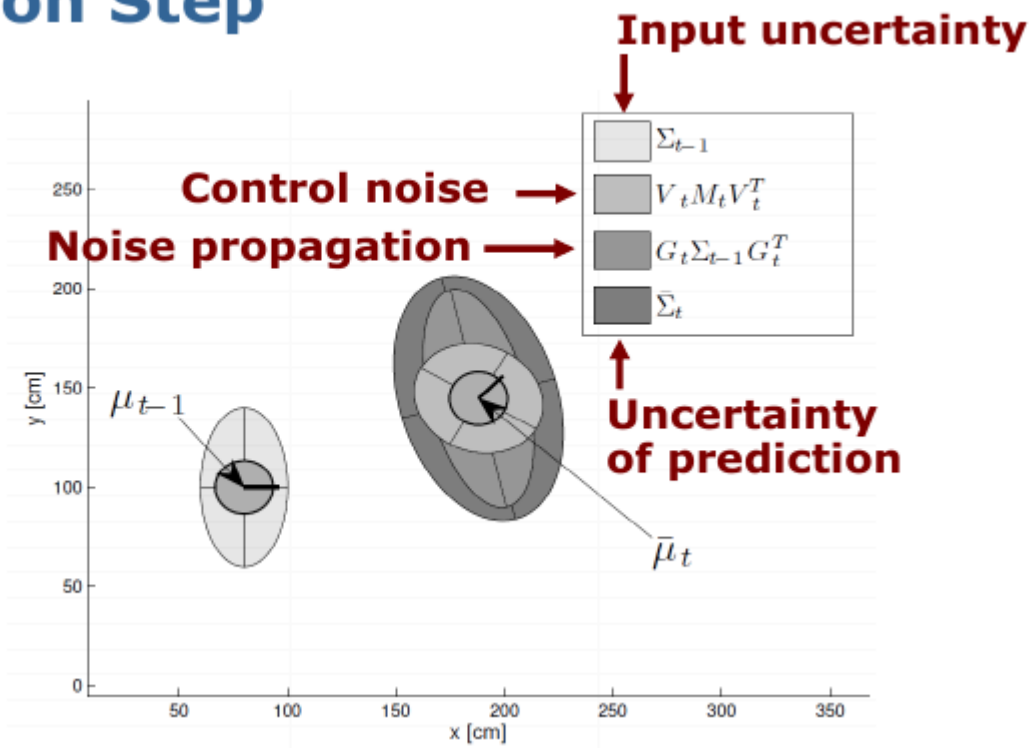
19: $\mu_t = \bar{\mu}_t$

20: $\Sigma_t = \bar{\Sigma}_t$

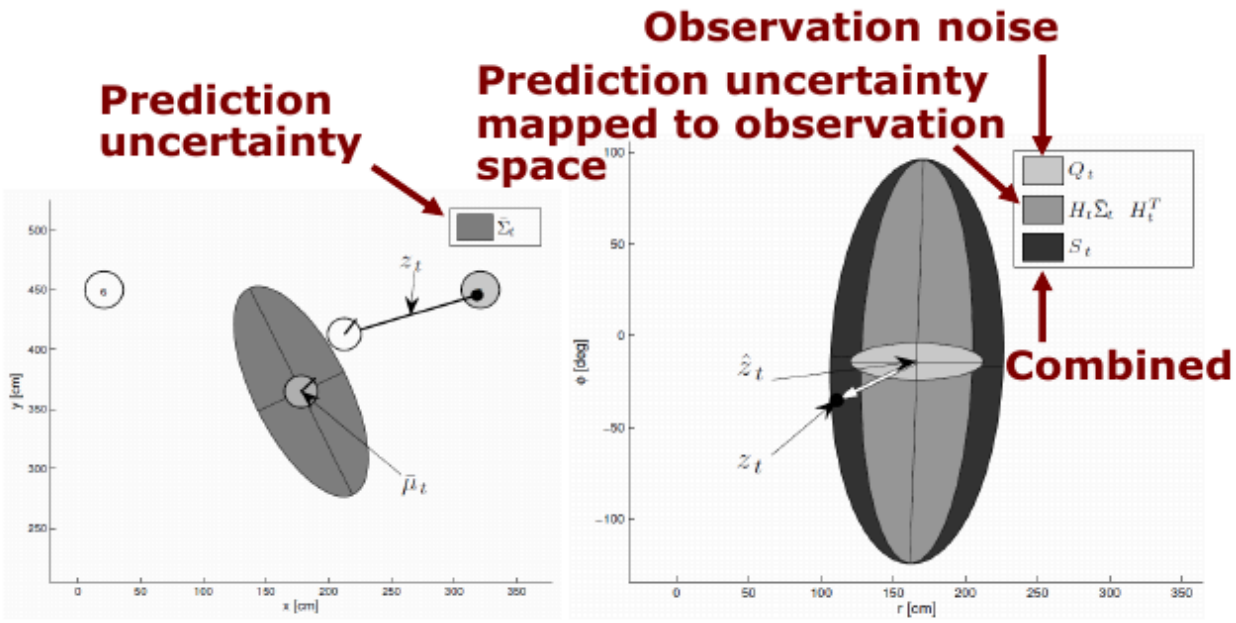
21: **return** μ_t, Σ_t

16 & 17 is Correction

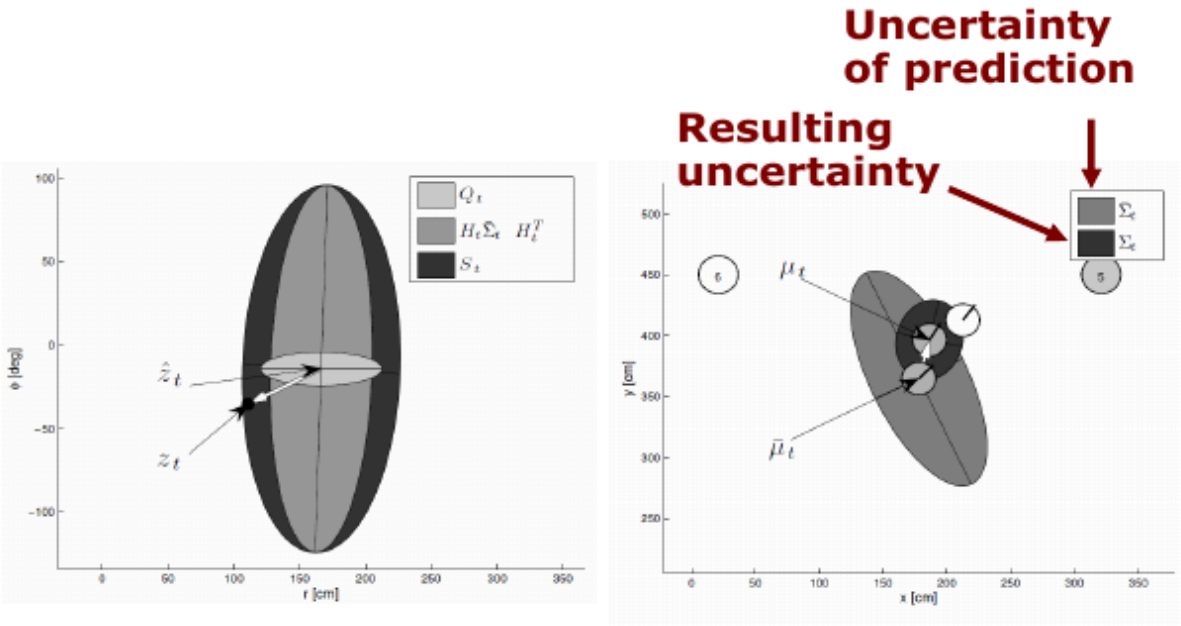
Prediction Step



Expected Measurement



Correction Step



EKF Localisation Summary

- EKF for pose estimation in the plane
- State vector: 3DoF pose
- Prediction using wheel odometry
- Correction using landmark observations
- Concrete example for both models and Jacobians